Status	Finished
Started	Wednesday, 21 October 2026, 3:51 PM
Completed	Wednesday, 21 October 2026, 3:56 PM
Duration	4 mins 57 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Bubble Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It sorts data using a single pass
- b. It does not require recursion
- c. It is based on the divide-and-conquer approach
- d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **5**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data sequentially
- c. Rearranging data without sorting
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **6**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It bubbles up the largest element in each pass
c. It divides the list into sublists
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sorted()
c. sort()
d. order()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **9**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Distribution
- c. Extraction
- d. Selection

The correct answer is:
Insertion

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Arranging data in a specific order
- c. Finding a specific element in a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It builds a sorted array one element at a time
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are compared element by element
c. They are split again into smaller sublists
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **14**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It allows for quicker identification of duplicates
c. It has no effect on the analysis process
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **15**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It slows down the process
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items